VALERIE JERUTO KOMEN

Data Analyst | SQL · Python · Power BI · Excel · Machine Learning

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SUMMARY

Data analyst with a civil engineering background, combining field-tested quantitative rigor (cost estimation, quality control, project reporting) with hands-on SQL, Python, and machine learning project work. Built four end-to-end analytics projects — data cleaning through predictive modeling to interactive dashboards — with a consistent focus on honest, defensible results over inflated ones.

TECHNICAL SKILLS

- **Languages & Tools:** SQL, Python (pandas, scikit-learn, matplotlib, seaborn), Excel (PivotTables, dashboards), Power BI / Plotly interactive dashboards, Git & GitHub
- **Techniques:** Data cleaning & multi-source reconciliation, exploratory data analysis, classification & regression modeling, feature engineering, statistical correlation analysis, data visualization, cross-validation

DATA ANALYTICS PROJECTS

- **Coffee Sales Analytics & ML** — github.com/Valerie-komen/Coffee-Sales-Analytics-ML
- End-to-end SQL/Python/ML pipeline on 1,000 transactions; built K-Means customer segmentation (silhouette score 0.56) and reported a repeat-purchase classifier's real limitations rather than an inflated accuracy figure; delivered an interactive Plotly dashboard.
- **Maji Ndogo Water Access Analytics** — github.com/Valerie-komen/Maji-Ndogo-Water-Access-Analytics
- Reconciled four inconsistent data exports covering 27.6M people into one validated SQL database; found only 46% basic water access nationally; built a well-contamination risk model (ROC-AUC 0.66) after catching and fixing a cross-validation methodology bug.
- **Maji Ndogo Farm Yield Prediction** — github.com/Valerie-komen/Maji-Ndogo-Farm-Yield-Prediction
- Found and corrected two data-quality bugs before modeling; parsed 10 free-text weather-message formats (English/Chinese) via regex at 100% coverage; predicted crop yield per hectare with Gradient Boosting ($R^2 = 0.96$, cross-validated).
- **Telco Customer Churn Analysis** — github.com/Valerie-komen/Telco-Customer-Churn-Analysis
- Identified contract length and payment method as the leading churn drivers (42.7% vs. 2.8% churn across contract types); built a classifier with stable cross-validated ROC-AUC of 0.845.

PROFESSIONAL EXPERIENCE

Assistant Engineer — Kenya National Highway Authority, Nairobi
July 2023 – Present

- Collected and analyzed road inventory data to support program-of-works planning and cost estimation.
- Built and maintained project progress dashboards and prepared monthly/quarterly reports for stakeholders.
- Verified work quantities and cost estimates, and reviewed contractor programmes against contractual timelines.
- Redesigned a road-base plan around an existing water pipeline, avoiding relocation costs and saving approximately KES 5 million.
- Introduced a daily reporting system that improved coordination between the Resident Engineer and contractor.

EDUCATION

Bachelor of Engineering, Civil & Structural Engineering
Moi University, Kenya

ALX Data Science Program — *In Progress (September 2026])*
Python, Professional Foundations, Data Analytics and Machine Learning.

AWS Cloud Practitioner Essentials — *In Progress (Expected August 2026)*
Amazon Web Services, via AWS Skill Builder